UNIVERSITY of HOUSTON | ECE

# ECE 3337: Signals & Systems Analysis

## Lecture 8: Linear Time Invariant Systems

Reference: Chapter 2
(Sections 2-6 and 2-7)

**Please turn off your gadgets now**

# Math Moment: The Triangle Inequality

$$|a + b| \leq |a| + |b|$$

**Works for all real numbers**
**Also works for integrals, for example:**

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)\, x(t-\tau)|\, d\tau$$

# New Concept: System Stability

Bounded Input → LTI System → Is the output Bounded or Infinite?

**As engineers, we are always interested in building stable systems (basically, systems that do not blow up!)**

**Basic Question: Can the output of the system go to infinity (unbounded) when a bounded input signal is applied?**

# Bounded Signals

**A signal is said to be bounded if there is a constant and finite upper bound $C$ so that:**

$$|x(t)| \leq C \text{ for all } t$$

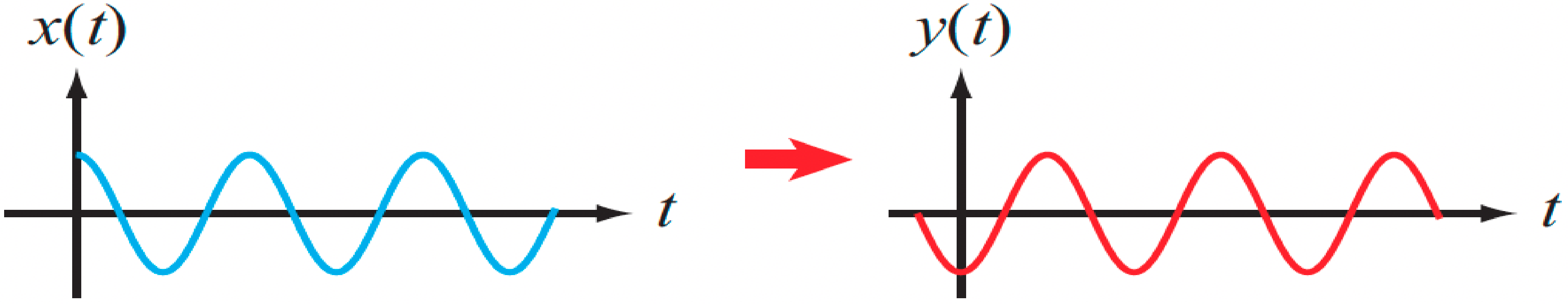
**Examples of Bounded Signals (what are the bounds?):**

$$\cos(3t), \quad 7e^{-2t}u(t), \quad e^{2t}u(1-t)$$

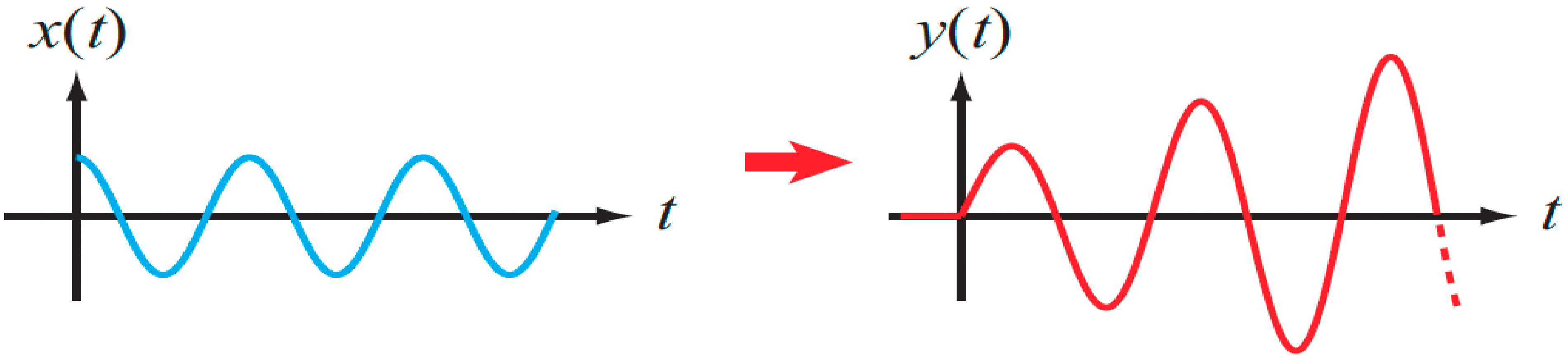
**Examples of Unbounded Signals**

$$t^2, \quad e^{2t}u(t), \quad e^{-t}, \quad 1/t$$

# Example

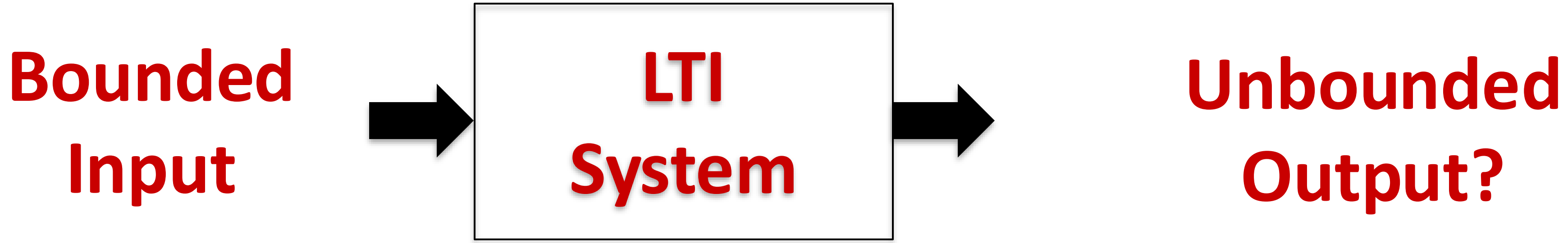



(a) BIBO-stable system




(b) Non-BIBO-stable system

# How is That Even Possible?




A bounded input can indeed result in an unbounded output

- Intuitively, this can happen when the energy in the bounded signal accumulates over time to create an unbounded output
  - E.g., a unit step or a sinusoidal input is bounded but carries infinite energy, so there is potential for an unbounded output
- It can also happen when the output contains momentary spikes (delta functions)

# BIBO Stability of a System

Bounded Input → LTI System → Is the output Bounded?

**Stability in the Bounded Input Bounded Output (BIBO) sense:**

- If the output is bounded (i.e., has a finite upper bound for <u>every</u> bounded input signal (no exceptions!), the system is said to be <u>BIBO Stable</u>
- Conversely, if the system generates an unbounded output for even one input waveform, it is not considered BIBO stable

# Example of BIBO Unstable System

$$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$

**INTEGRATOR**

**Let's try the bounded "test" input $x(t) = u(t)$, then**

$$y(t) = \int_{-\infty}^{t} u(\tau) d\tau = \int_{0}^{t} u(\tau) d\tau = t u(t)$$

**THIS IS UNBOUNDED**
**$\Rightarrow$ NOT BIBO STABLE**

**Intuitively, systems that endlessly accumulate signal energy over time are often BIBO unstable**

# Another Example of BIBO Unstable System

$$y(t) = \frac{dx(t)}{dt}$$

**Let's try the bounded "test" input $x(t) = u(t)$, then**

$$y(t) = \delta(t)$$

**THIS IS UNBOUNDED**
**$\Rightarrow$ NOT BIBO STABLE**

**Abruptly changing (bounded) test signals can often reveal BIBO Instability**

# BIBO Stability Testing

$$x(t) = \delta(t)$$ × (UNBOUNDED)

$$\boxed{\begin{aligned} x(t) &= u(t) \\ x(t) &= \cos(\omega t) \end{aligned}}$$

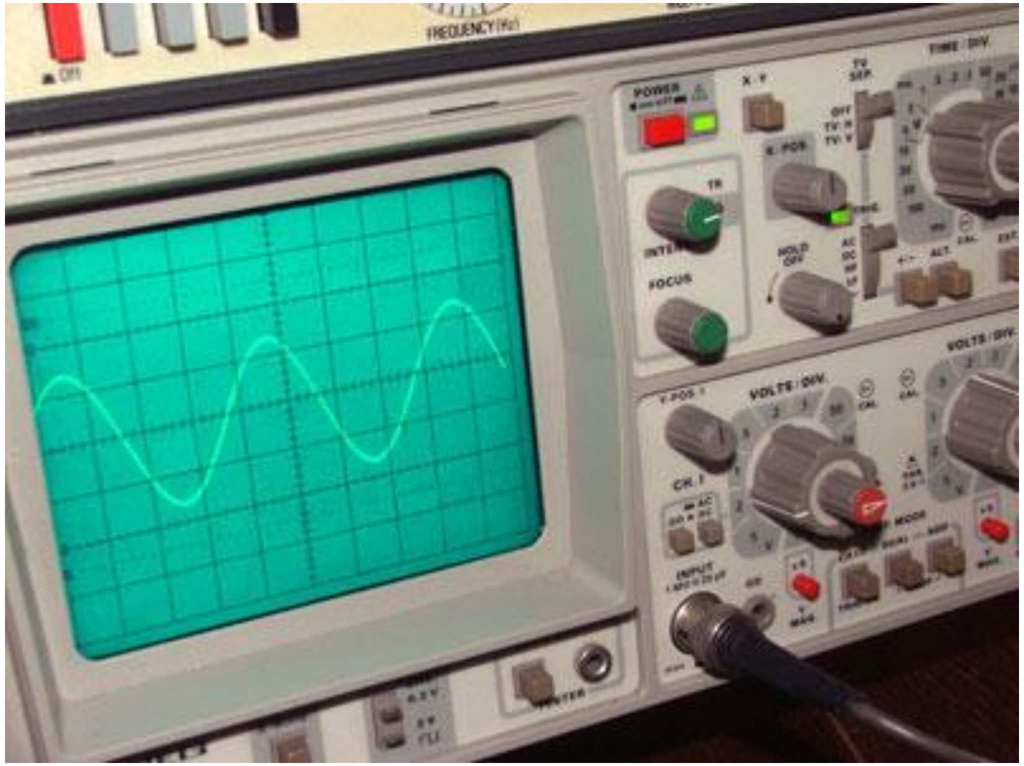
## The unit step and sinusoid are great signals for BIBO Stability Testing

- They are finite in terms of amplitude, but carry an infinite amount of energy that can accumulate over time to create an unbounded output

- The unit step also has an abrupt change at $t = 0$

# Common Test Signals

**Input** $x(t)$ → **LTI System** → **Response** $y(t)$



$$x(t)=\delta(t)$$ × (UNBOUNDED)

$$x(t)=u(t)$$

$$x(t)=\cos(\omega t)$$

**LET US GET TO KNOW THIS ONE TODAY**

# Analyzing the Impulse Response

**Since the impulse response $h(t)$ is a complete description of an LTI system, it is helpful to be able to infer BIBO stability from $h(t)$**

$$|x(t)| \leq M$$ **BOUNDED**

$$\int_{-\infty}^{\infty} |h(t)|\, dt = L < \infty$$ **BOUNDED**

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)\, x(t-\tau)|\, d\tau$$

**TRIANGLE INEQUALITY**

$$\leq \int_{-\infty}^{\infty} |h(\tau)| M\, d\tau = LM$$

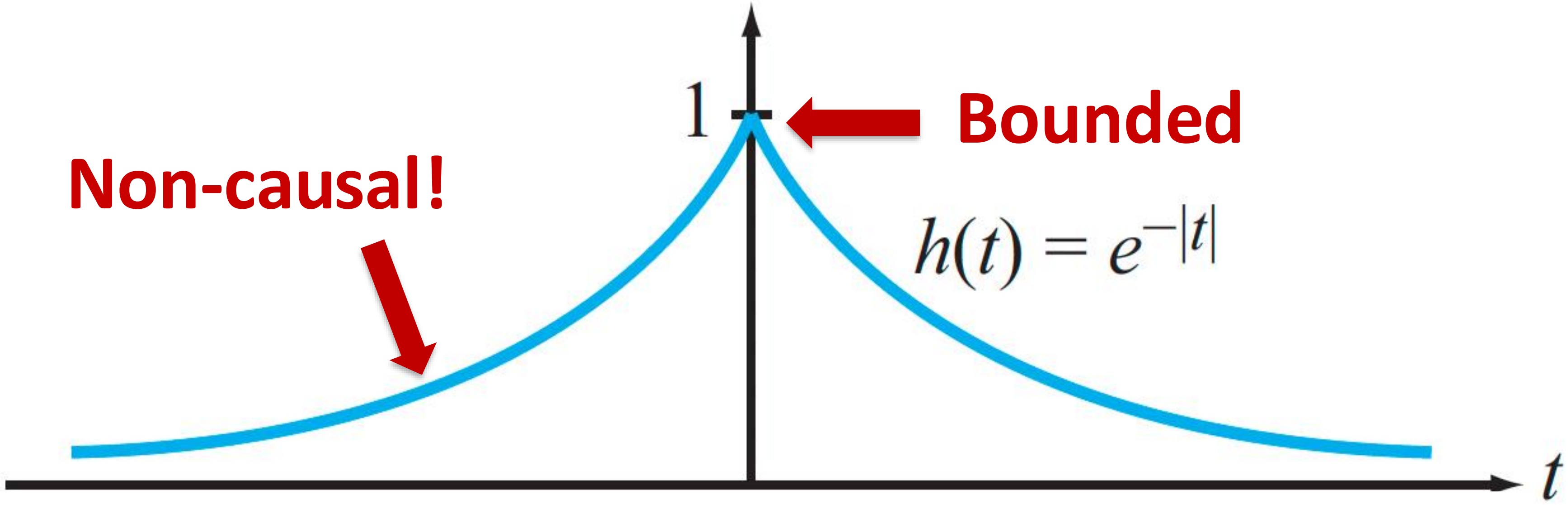
**BOUNDED**

# General Theorem

**An LTI system is BIBO stable if and only if the impulse response is "absolutely integrable":**

$$\int_{-\infty}^{\infty} |h(t)|\, dt \text{ is finite}$$

# Example




$$\int_{-\infty}^{\infty} |h(t)|\, dt = \int_{-\infty}^{\infty} e^{-|t|}\, dt = 2\int_{0}^{\infty} e^{-t}\, dt = 2.$$

**FINITE**

Hence, the system is BIBO stable.

# Watch the Exponentials

**Exponential impulse responses are an important case**

$$h(t) = e^{-at}u(t), a > 0$$ **Causal, BIBO stable**

$$h(t) = e^{at}u(t), a > 0$$ **Causal, Not BIBO stable!**

$$h(t) = e^{j\omega t}u(t)$$ **Causal, Not BIBO stable!**

$$h(t) = e^{at}u(t), a = 0$$ **Causal, Not BIBO stable!**

**<u>Bottom Line</u>: ONLY the decaying exponential case is BIBO stable**

# LTI Response to Sinusoids

**What happens when we apply a sinusoidal input to a LTI system?**

**It is easiest to start with the exponential form:**

$$x(t) = e^{j\omega t} \longrightarrow \boxed{\begin{array}{c}\text{LTI System}\\ h(t)\end{array}} \longrightarrow y(t) = e^{j\omega t} * h(t)$$

**Where $\omega$ is the angular frequency in radians per second**
**The exponential form of the sinusoid $e^{j\omega t}$ is mathematically easier to work with compared to $\cos(\omega t)$ or $sin(\omega t)$**

**Remember Euler's Formula: $e^{j\omega t} = cos(\omega t) + jsin(\omega t)$**

# LTI Response to Sinusoids

$$x(t) = e^{j\omega t} \rightarrow \boxed{\text{LTI System } h(t)} \rightarrow y(t) = e^{j\omega t} * h(t)$$

$$y(t) = e^{j\omega t} * h(t) = \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau$$

$$= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau = e^{j\omega t} \widehat{\mathbf{H}}(\omega)$$

$$\widehat{\mathbf{H}}(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

**The Frequency Response Function**

# Practical Meaning of the Frequency Response

## The Frequency Response is a complex quantity, even if $h(t)$ is real

$$e^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow \underbrace{\hat{\mathrm{H}}(\omega)}_{\text{ADJUSTMENT}}\, e^{j\omega t}$$

<u>**Important observations:**</u>

1. **The output is also a sinusoid at the same exact frequency $\omega$**
2. **The amplitude of the output is adjusted by $\left|\widehat{H}(\omega)\right|$, a frequency dependent quantity – magnitude spectrum**
3. **The phase of the output is adjusted by $\theta = \angle \hat{\mathrm{H}}(\omega)$, also a frequency dependent quantity – phase spectrum**

# Valuable Observations

$\hat{\mathbf{H}}(\omega)$ is independent of $t$

$\hat{\mathbf{H}}(\omega)$ is a function of $j\omega$, not just $\omega$

$\hat{\mathbf{H}}(\omega)$ is defined for all values of $t$ from $-\times$ to $\times$

If $h(t)$ is real, then,

$\hat{\mathbf{H}}(-\omega) = \hat{\mathbf{H}}^{*}(\omega)$ because

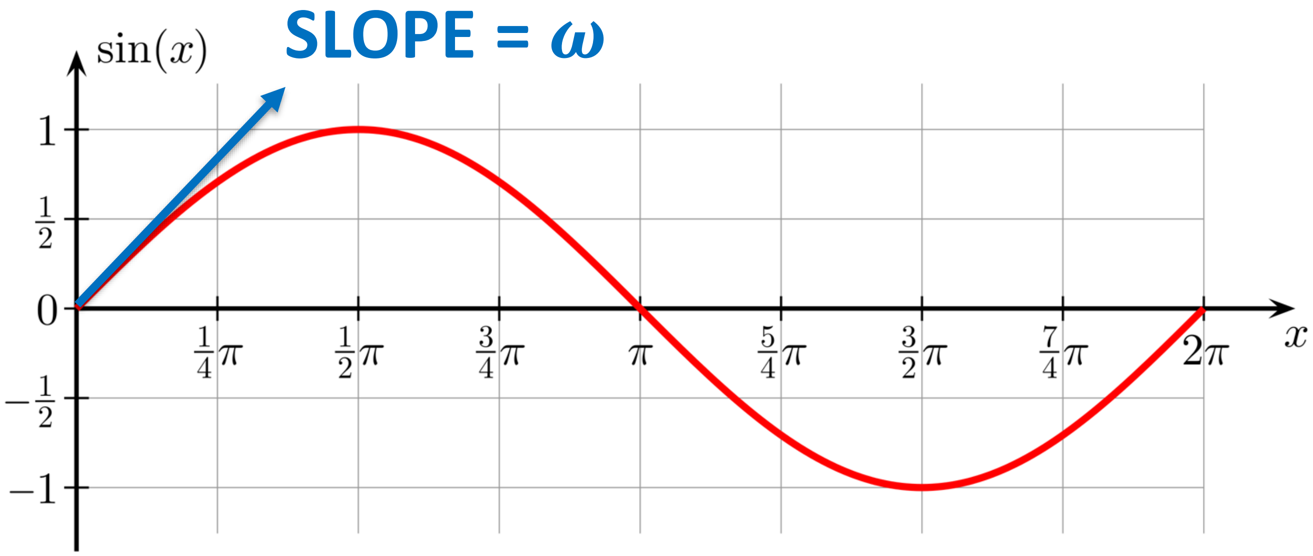
$$\hat{\mathbf{H}}^{*}(\omega) = \square_{-\times}^{\times} h^{*}(\tau)[e^{-j\omega t}]^{*} d\tau = \square_{-\times}^{\times} h(\tau)e^{j\omega t} d\tau = \hat{\mathbf{H}}(-\omega)$$

**Conjugate Symmetry**

# Some Intuition

**For sinusoids, the concept of frequency is directly tied to the rate of change of the signal**

$$\frac{d}{dt}[\sin(\omega t)] = \omega \cos(\omega t)$$




**“HIGH FREQUENCY” SIGNALS ARE “FAST CHANGING”**

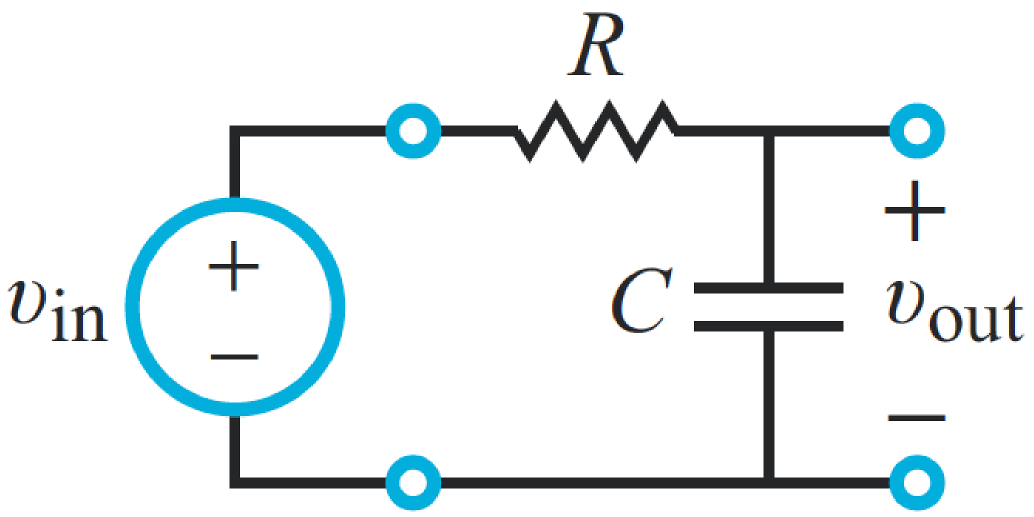
# RC Circuit Example

**Recall that when RC = 1,**

$$h(t) = e^{-t}u(t)$$

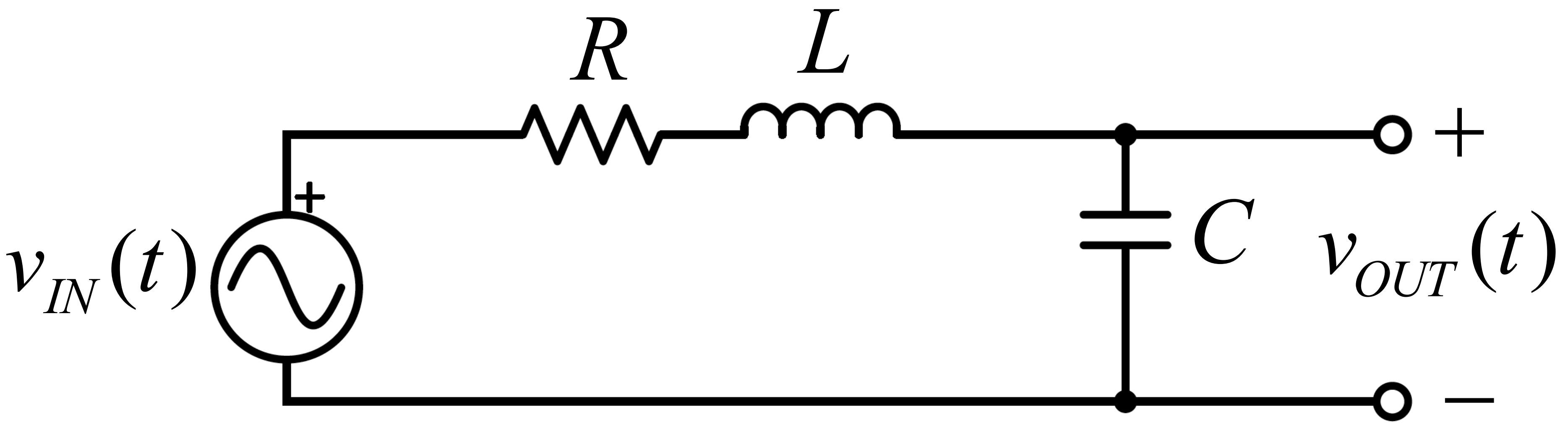
$$\hat{\mathrm{H}}(\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}\,d\tau$$

$$= \int_{0}^{\infty} e^{-\tau}e^{-j\omega\tau}\,d\tau = \int_{0}^{\infty} e^{-(1+j\omega)\tau}\,d\tau = \frac{1}{1+j\omega}$$




**THIS CIRCUIT ATTENUATES HIGHER FREQUENCIES BUT PASSES LOW FREQUENCIES – It is a "Low Pass" Filter**

# RLC Circuit Example




**Using KVL, we can show that this circuit behaves according to:**

$$v_{IN}(t) = LC\frac{d^2 v_{OUT}(t)}{dt^2} + RC\frac{dv_{OUT}(t)}{dt} + v_{OUT}(t)$$

**This is a Linear Constant Coefficient Differential Equation (LCCDE)**

# LCCDE's

## Linear constant-coefficient differential equations (LCCDE's) occur commonly when describing LTI systems

**Example:**

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 4\frac{dx}{dt} + 5x$$

- **These systems are always LTI**
- **For such systems, it is particularly convenient to find the response to a sinusoidal input**
- **It is also very practical, since sinusoids are common and easy to generate**

# Biomedical LCCDE Example

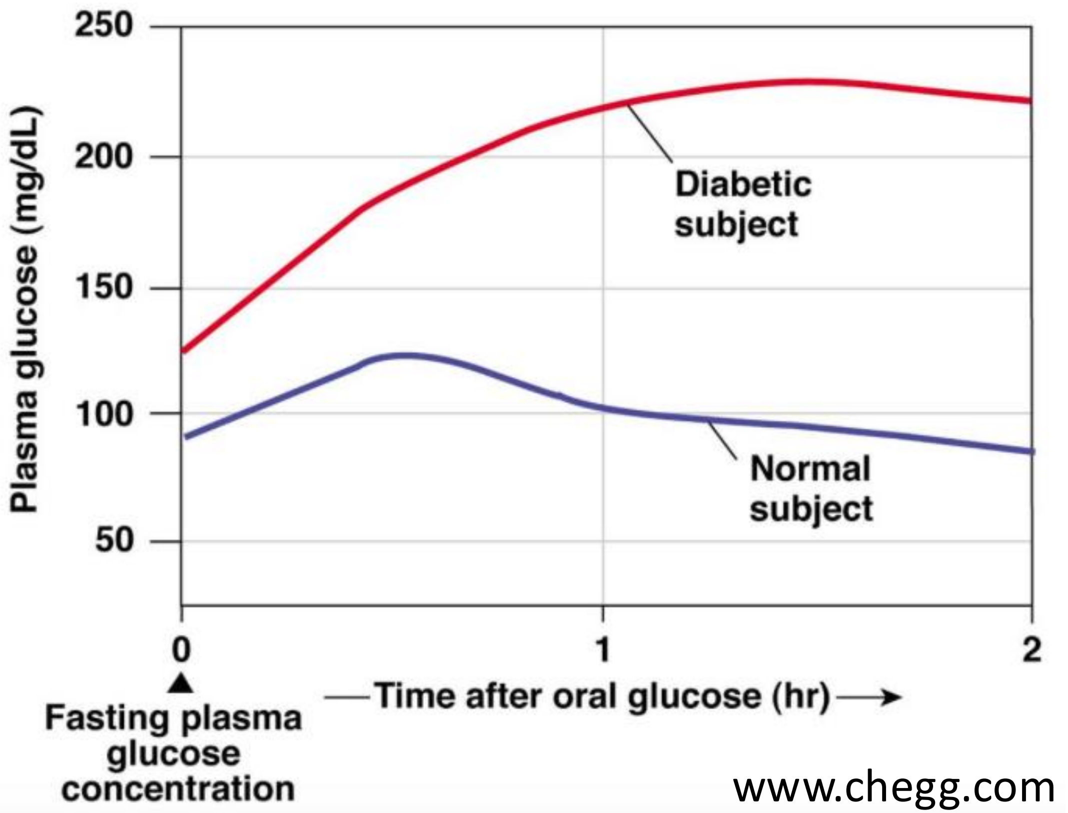
In a glucose tolerance test (GTT), a patient's baseline blood glucose is measured, and then he/she is given a bolus (~75g) of glucose (~ an impulse)

- The patient's blood glucose concentration above the baseline value, $g(t)$ can be modeled using a 2nd order LCCDE
- The patient-dependent constants $\beta$ $\omega_0$ can be used for diagnosis, and for controlling medication delivery

$$\frac{d^2 g(t)}{dt^2} + \beta \frac{dg(t)}{dt} + \omega_0^2 g(t) = Q(t)$$

Patient-dependent constants

Glucose Input

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# Example (contd)

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 4\frac{dx}{dt} + 5x$$

$$x(t) = e^{j\omega t}$$

$$y(t) = \hat{\mathbf{H}}(\omega)e^{j\omega t}$$

**Keep in mind that:**

$\hat{\mathbf{H}}(\omega)$ is independent of $t$

$$\hat{\mathbf{H}}(\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}\,d\tau$$

# Example (contd)

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 4\frac{dx}{dt} + 5x$$

$$x(t) = e^{j\omega t}$$

$$y(t) = \hat{\mathbf{H}}(\omega)e^{j\omega t}$$

$$\frac{dx}{dt} = j\omega e^{j\omega t}$$

$$\frac{dy}{dt} = j\omega\hat{\mathbf{H}}(\omega)e^{j\omega t}$$

$$\frac{d^2y}{dt^2} = (j\omega)^2\hat{\mathbf{H}}(\omega)e^{j\omega t}$$

**Differentiating a signal in the time domain translates to a multiplication by $j\omega$ in the frequency domain**

# Example (contd)

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 4\frac{dx}{dt} + 5x$$

**Substituting:**

$$(j\omega)^2 \square \hat{\mathbf{H}}(\omega)e^{j\omega t} + 2j\omega \square \hat{\mathbf{H}}(\omega)e^{j\omega t} + 3\hat{\mathbf{H}}(\omega)e^{j\omega t} = 4j\omega e^{j\omega t} + 5e^{j\omega t}$$

**Solve for** $$\hat{\mathbf{H}}(\omega) = \frac{5 + 4j\omega}{-\omega^2 + 2j\omega + 3}$$

**$\hat{H}(\omega)$ IS A COMPLETE SPECIFICATION OF THE LTI SYSTEM, JUST LIKE $h(t)$**

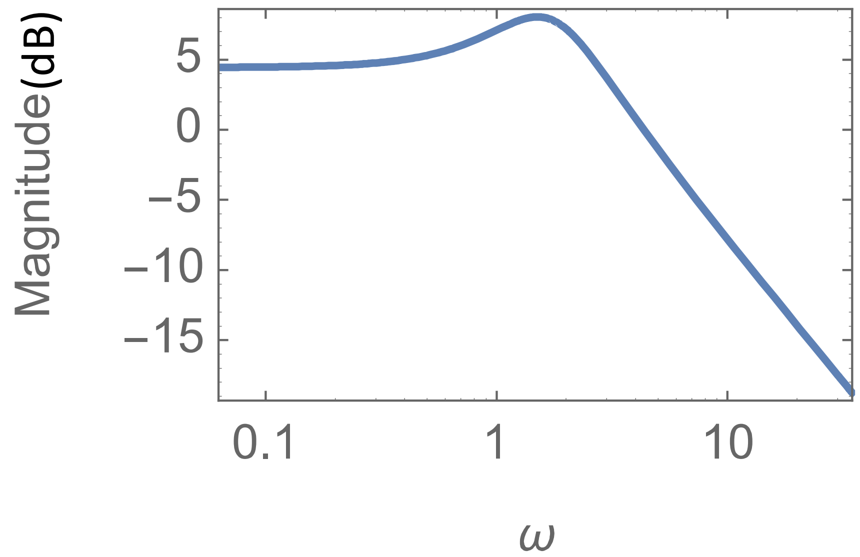
# Example (contd)

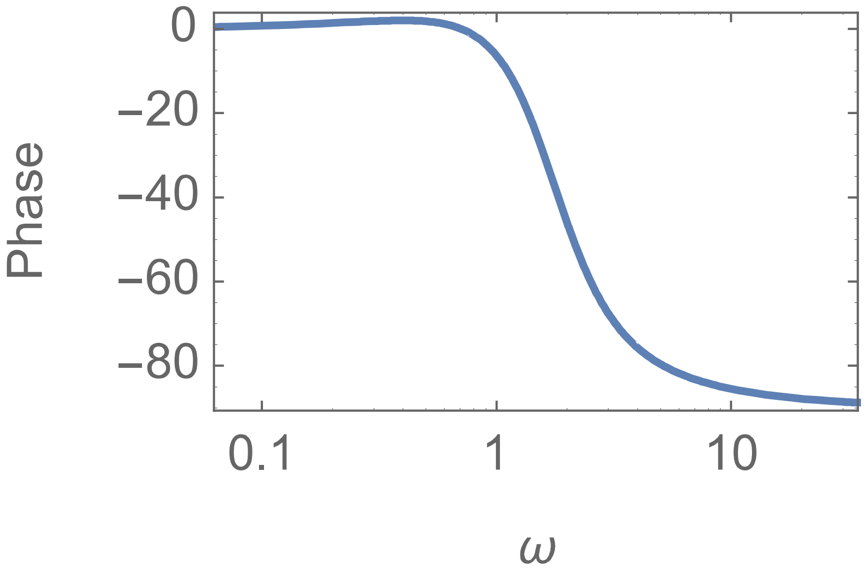
$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 4\frac{dx}{dt} + 5x$$

$$\hat{\mathbf{H}}(\omega) = \frac{5 + 4j\omega}{-\omega^2 + 2j\omega + 3}$$

**HOW DOES THIS VARY WITH THE FREQUENCY $\omega$?**

- **MAGNITUDE PLOT**
- **PHASE PLOT**



# Try the Following Example

$$3\frac{d^3y}{dt^3}+6\frac{dy}{dt}+4y=5\frac{dx}{dt}$$

Set: $x(t)=e^{j\omega t}$

$$y(t)=\hat{\mathbf{H}}(\omega)e^{j\omega t}\text{, and solve for }\hat{\mathbf{H}}(\omega)$$

Correct Answer:

$$\hat{\mathbf{H}}(\omega)=\frac{5(j\omega)}{4+6(j\omega)+3(j\omega)^3}$$

# Working Backwards

**Given:** $$\hat{\boldsymbol{H}}(\omega) = \frac{5(j\omega)}{3(j\omega)^3+6(j\omega)+4}$$

**We can guess the differential equation that it came from, by simply working backwards:**

$$3(j\omega)^3 e^{j\omega t}\hat{\boldsymbol{H}}(\omega) + 6(j\omega)e^{j\omega t}\hat{\boldsymbol{H}}(\omega) + 4e^{j\omega t}\hat{\boldsymbol{H}}(\omega) = 5(j\omega)e^{j\omega t}$$

**Answer:** $$3\frac{d^3y}{dt^3} + 6\frac{dy}{dt} + 4y = 5\frac{dx}{dt}$$

# Recap: Euler's Formula

***e*** **is called the "Euler's Number"**

$$e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$$

$$e^{-j\omega_0 t} = \cos(\omega_0 t) - j\sin(\omega_0 t)$$

**Realizing just how common sinusoids are, it makes sense to also look at the following result:**

$$e^{-j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow \hat{\mathrm{H}}(-\omega)\, e^{-j\omega t}$$

# LTI Response to Sinusoids

$$x(t) = \cos \omega t = \frac{1}{2}[e^{j\omega t} + e^{-j\omega t}]$$

**Using the Additivity property of LTI systems, we see that:**

$$\frac{1}{2} e^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow \frac{1}{2} \hat{\mathbf{H}}(\omega)\, e^{j\omega t}$$

$$+ \qquad\qquad\qquad\qquad +$$

$$\frac{1}{2} e^{-j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow \frac{1}{2} \hat{\mathbf{H}}(-\omega)\, e^{-j\omega t}.$$

# Observation

$$\hat{\mathbf{H}}(-\omega) = \hat{\mathbf{H}}^*(\omega).$$

Complex Conjugate

$$\hat{\mathbf{H}}(\omega) = |\hat{\mathbf{H}}(\omega)|e^{j\theta},$$

$$\hat{\mathbf{H}}(-\omega) = \underbrace{|\hat{\mathbf{H}}(\omega)|}_{\text{Magnitude (unchanged)}}\underbrace{e^{-j\theta}}_{\text{Phase (flipped)}},$$

ONLY THE PHASE CHANGES

# Cosine Response of LTI Systems

$$y(t)=\frac{1}{2}\left[\hat{\boldsymbol{H}}(\omega)e^{j\omega t}+\hat{\boldsymbol{H}}(-\omega)e^{-j\omega t}\right]$$

$$y(t)=\frac{1}{2}\left|\hat{\boldsymbol{H}}(\omega)\right|\left[e^{j(\omega t+\theta)}+e^{-j(\omega t+\theta)}\right]$$

$$y(t)=\left|\hat{\boldsymbol{H}}(\omega)\right|\cos\left(\omega t+\theta\right)$$

**<u>Bottom line</u>: the LTI system response to a cosine function is the magnitude of the transfer function multiplied by a cosine of the same frequency but a different phase (delay).**

# The Cosine Response of LTI Systems

**SINUSOID IN**

**SINUSOID OUT @ SAME FREQUENCY**

$$\cos \omega t \longrightarrow \boxed{h(t)} \longrightarrow \underbrace{|\mathbf{H}(\omega)|}_{\text{Amplitude Change}} \cos(\omega t + \underbrace{\theta}_{\text{Phase Change}})$$

$$\theta = \angle \hat{\mathbf{H}}(\omega)$$

# Important Observation

Note that the response of an LTI system to a sinusoid is at the same frequency as the input

- Only the magnitude and phase are altered

- Corollary:
  - If a system outputs any new frequencies that are not in the input, then the system is not linear, so no longer LTI
  - Such systems are said to be “non-linear”
    - Generally messy to work with, advanced math needed
    - Can often be approximated with a linear system
    - Non-linear systems are beyond the scope of this course

# Example #1

## Observed LTI system behavior:

$$x(t) = 1 + 2\cos(t) + 3\cos(2t)$$

$$y(t) = 6\cos(t) + 6\cos(2t - 45^o)$$

$$\hat{\mathbf{H}}(0) = \frac{0}{1} = 0$$

$\boldsymbol{\omega} = \mathbf{0},$
**PHASE = 0º**

$$\hat{\mathbf{H}}(1) = \frac{6}{2} = 3$$

$\boldsymbol{\omega} = \mathbf{1},$
**PHASE = 0º**

$$\hat{\mathbf{H}}(2) = \frac{6}{3}e^{-j45^o} = 2e^{-j45^o}$$

$\boldsymbol{\omega} = \mathbf{2},$
**PHASE = - 45º**

**WE CANNOT INFER ANYTHING ABOUT ANY OTHER FREQUENCIES FROM THE GIVEN DATA**

# Example #1 (contd)

**Calculate the response of the same LTI system to the following new input:**

$$x(t) = 7 + \cos(2t)$$

Correct Answer:

$$\mathrm{y}(t) = 0 + 2\cos(2t - 45^{o})$$

# Example #2

**Find the output of the following system when $\omega = 4$**

$$\frac{dy}{dt} + 3y = 3\frac{dx}{dt} + 5x$$

$$\text{Input: } x(t) = 5\cos(4t + 30^{o})$$

**Since the input is a sinusoid, it is easier to work with the frequency response function rather than $h(t)$**

**Step 1: Calculate the frequency response function**

$$\hat{\mathbf{H}}(\omega) = \frac{5 + 3(j\omega)}{3 + (j\omega)}$$

# Example #2

$$\hat{\mathbf{H}}(\omega) = \frac{5 + 3(j\omega)}{3 + (j\omega)}$$

$$\boxed{\begin{array}{l} \dfrac{dy}{dt} + 3y = 3\dfrac{dx}{dt} + 5x \\ \text{Input: } x(t) = 5\cos(4t + 30^O) \end{array}}$$

**Step 2: Calculate the response when** $\omega = 4$

$$\hat{\mathbf{H}}(4) = \frac{5 + 3(j4)}{3 + (j4)} = \frac{13e^{j67^o}}{5e^{j53^o}} = \frac{13}{5}e^{j14^o}$$

$$y(t) = 5 \cdot \frac{13}{5}\cos(4t + 30^O + 14^O)$$

# Summary: LTI Response to Sinusoids

**SINUSOID IN:**

$$x(t) = A\cos(\omega t + \phi) \rightarrow \boxed{h(t)} \rightarrow y(t) = A\underbrace{\left|\hat{\mathbf{H}}(\omega)\right|}_{\text{Amplitude Change}}\cos(\omega t + \phi + \underbrace{\theta}_{\text{Phase Change}})$$

$$\theta = \angle\hat{\mathbf{H}}(\omega)$$

$$\boxed{\hat{\mathbf{H}}(\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau}$$

Frequency-domain version of $h(t)$

**SAME FREQUENCY SINUSOID OUT,**

- **CHANGED AMPLITUDE**
- **CHANGED PHASE**

# This is a Big Deal

When we get to Chapter 5 (Fourier Transform), we will use this insight to analyze complicated signals

- Complicated signals can be re-written as a sum of sinusoids of different frequencies
- We can apply this result to each of the sinusoids individually, and add up the responses to calculate the overall response
- In the lab, we can characterize systems by their frequency response using sinusoidal inputs. This is way more convenient than the impulse or step responses

# For the Next Class

**Work through the textbook examples in Sections 2.6 – 2.7**

**Study Section 2.8 *on Zybooks***

How to Study:

1. **Read the assigned sections to understand the basic concepts, and memorize them**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**